Optical Transmission Measurements

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1 Introduction

The purpose of this experiment was to determine the optical transmittance of two Kurt J. Lesker CF Flanged Quartz (DUV Fused Silica) Viewports, two Kurt J. Lesker CF Flanged Exotic Lens Calcium Fluoride Viewports, an unknown viewport mounted on a 6" flange, and an unknown viewport mounted on a 9" inch flange.

2 Methods and Procedure

2.1 Materials

- Deuterium, Hydrogen, and Neon lamp
- StellerNet spectrometer
- SpectraWiz spectrometer software
- 3-D printed parts
  - (2) Lamp couplers
  - (2) Optical fibre couplers
  - (4) Viewport supports
- Black cloth

2.2 Experimental Procedure

The experiment was set up so that light from the lamp first passes through the lamp coupler, which isolates light from one section of the lamp. The isolated light travels through the lamp coupler's cylindrical passage which connects directly to the viewport. The viewport is mounted on a support which aligns it with the lamp coupler. After passing through the viewport, light travels through the optical fibre coupler, a cylindrical part with a nozzle at the end where the optical fibre is inserted. The entire setup is mounted on an optical table and covered with a large black cloth to minimize background light.

Two sets of couplers were designed: one to fit the Calcium Fluoride (CaF_2) viewports (10.5mm radius) and one to fit the DUV fused silica (DUV) viewports (17mm radius). The CaF_2 couplers connected perfectly, but due to the reverse side of the DUV window having a larger radius, the DUV optical fibre coupler left some area exposed. Measurements of the 6" and 9" flange viewports were taken using the CaF_2 couplers. Much of the 6" flange's window was exposed since it was very large. The 9" flange's window was only slightly larger than the CaF_2 coupler radius, so less was exposed.

The optical transmittance of each viewport was measured with three different light sources: a Deuterium (D), a Hydrogen (H), and a Neon (Ne) lamp. For each lamp-viewport combination, 10 measurements were taken using the

StellerNet spectrometer. Each measurement was taken one minute apart with a 1000ms integration time. The data was collected with the SpectraWiz software which provides information for the intensity of light at wavelengths between 184nm and 847.5nm.

3 Results and Discussion

3.1 Calculations

Python modules pyplot and matplotlib were used to complete calculations and generate graphs for the data.

First, the mean of each set of 10 measurements was calculated. This was done by summing the 10 arrays and obtaining the mean of each value in the array.

For the two CaF_2 and two DUV viewports, the mean of the two viewports was calculated such that if c_1 and c_2 were mean arrays of the first CaF_2 viewport and the second CaF_2 viewport, respectively, then the values obtained from $(c_1 + c_2)/2$ were used when moving forward with the calculations.

A base measurement of background light was obtained and subtracted from each mean array. Fresnel corrections were calculated for the CaF_2 and DUV viewports but were not included in the final result. Fresnel corrections could not be computed for the unknown viewports mounted in the 6" and 9" flanges.

The transmission spectra were obtained by computing *intensity of obstructed light / intensity of unobstructed light* where obstructed light was light that had passed through a viewport and unobstructed light was light directly from a lamp. Values less than 0 and values greater than 1 were ignored.

3.2 Plots

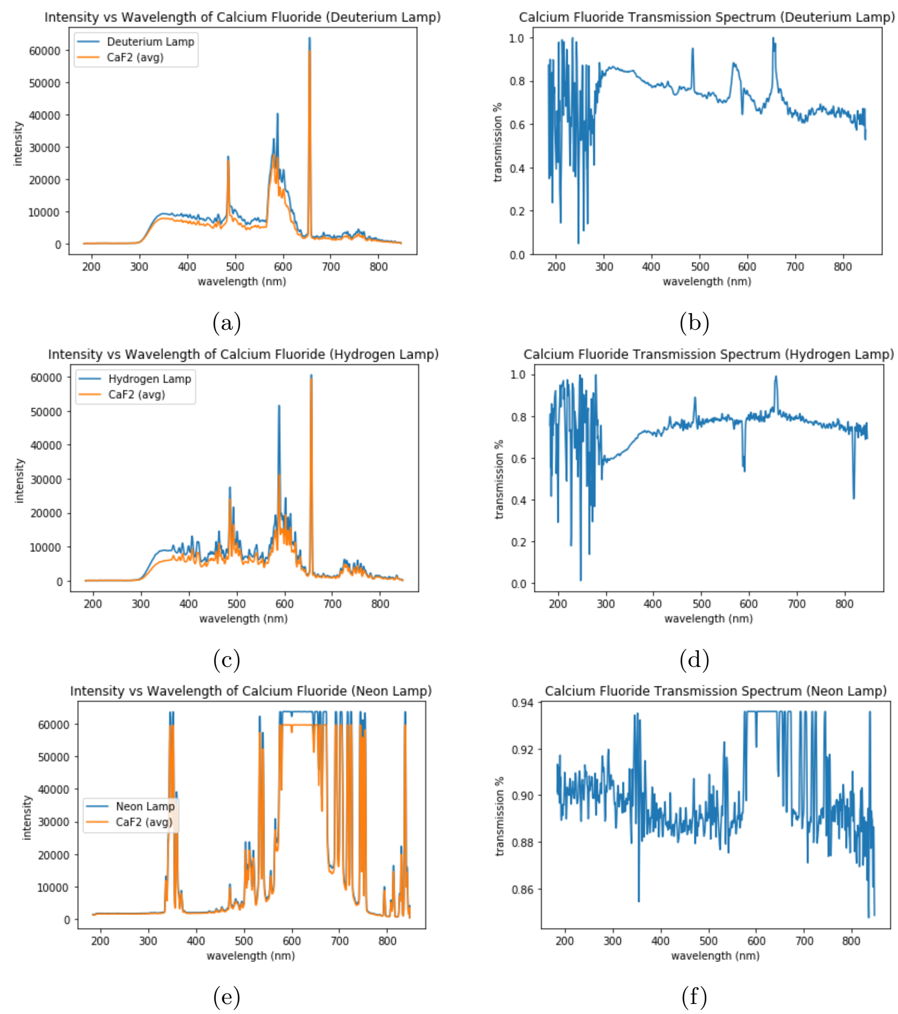
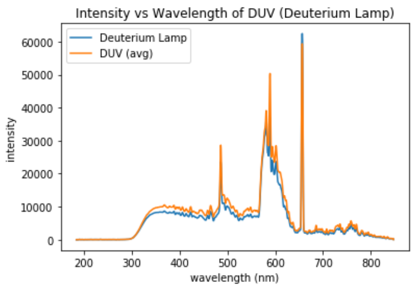


Figure 1: CaF_2 viewports with D, H, and Ne lamps.



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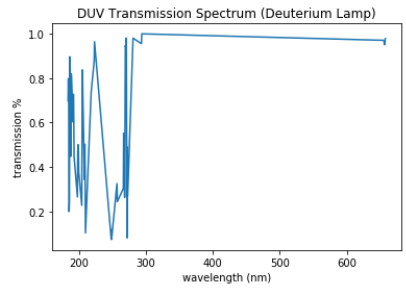

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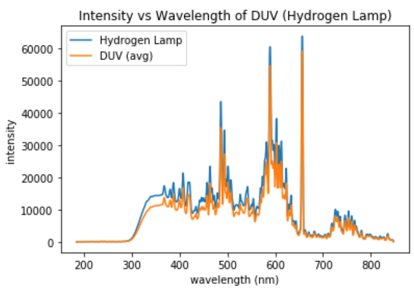

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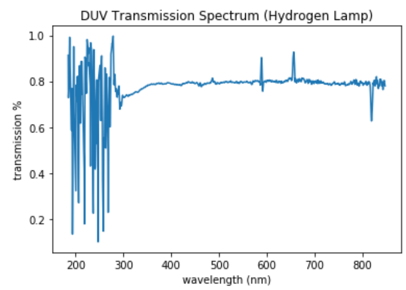

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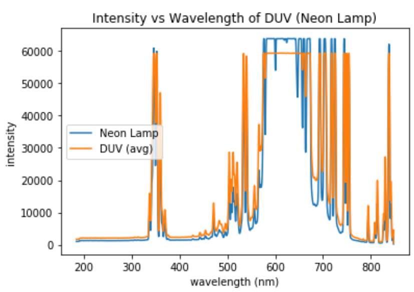

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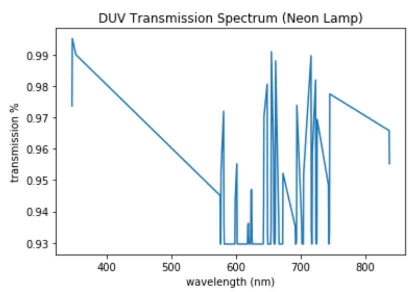
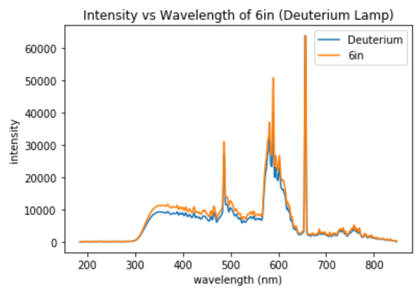
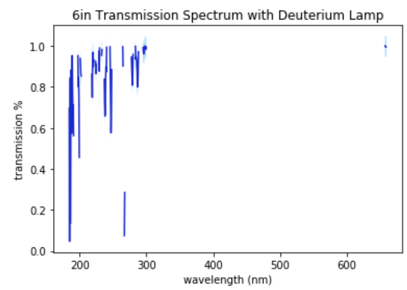


(f)

Figure 2: DUV viewports with D, H, and Ne lamps.



(a)



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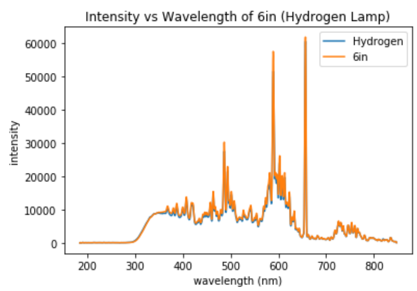

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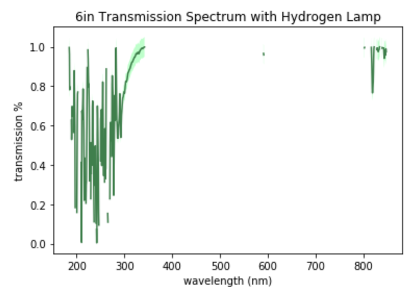

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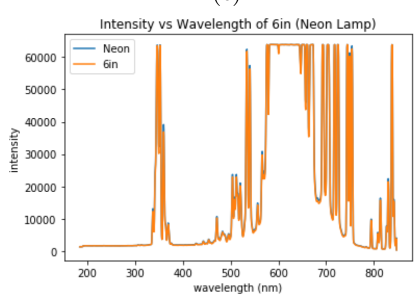

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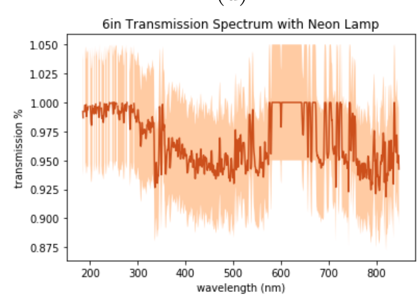
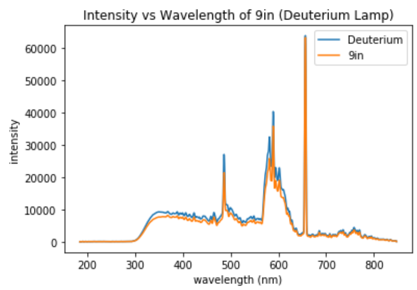
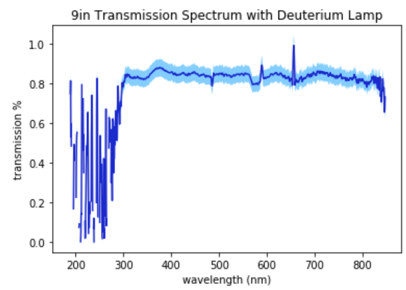


(f)

Figure 3: 6” flange with viewport with D, H, and Ne lamps.



(a)



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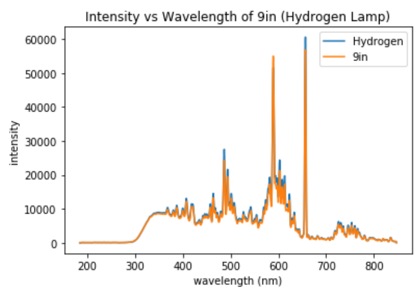

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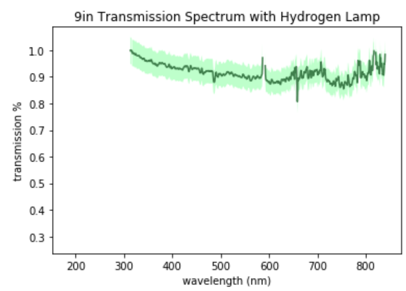

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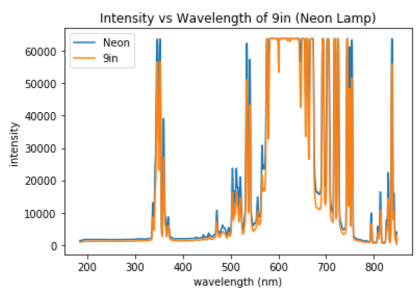

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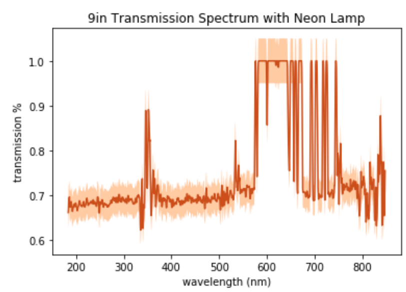

(f)

Figure 4: 9" flange with viewport with D, H, and Ne lamps.

3.3 Error Analysis

The error obtained was too small a value to be seen on the graphs. The shaded areas in the transmission spectra graphs in Figure 3 and Figure 4 have an assumed 5 percent error.

Measurements done with the Ne lamp created odd results for all of the viewports. This may be due to the intensity of the Ne lamp exceeding what the spectrometer can process in wavelengths from around 550 - 750nm.

In Figure 2a, the intensity of the obstructed light seems to exceed the intensity of the unobstructed light. A similar effect is especially prominent in Figure 3a, 3b, and 3c where much of the ratio of obstructed to unobstructed light exceeded 1 and was not plotted, resulting in a discontinuous transmission spectra. In Figure 4a, 4b, and 4c, the transmission spectra also shows a few areas of discontinuity. This phenomena may be caused by failure to account for the distance that the unobstructed light traveled in the absence of a viewport or since Fresnel corrections were not accounted for.

4 Conclusions

Better measurements may be obtained using custom coupling parts for each flange, in addition to a ring insert that would account for the thickness of the viewport through which light must travel when the viewport is absent. If the identities of the unknown viewports in the 6" and 9" flanges are discovered, it would be ideal to apply Fresnel corrections to the data.